7(a)(i)	$0.52 = \frac{N}{t}\left(1.6 \times 10^{-19}\right)$ $\frac{N}{t} = \frac{0.52}{1.6 \times 10^{-19}}$ $\frac{N}{t} = 3.25 \times 10^{18}\ \text{s}^{-1} \approx 3.3 \times 10^{18}\ \text{s}^{-1}$	[1] sub Must show $\frac{N}{t} = 3.25 \times 10^{18}$
7(a)(ii)	$v = \frac{0.52}{\left(5.9 \times 10^{28}\right)(\pi)\left(0.18 \times 10^{-3}\right)^2\left(1.60 \times 10^{-19}\right)}$ $= 5.4 \times 10^{-4}\ \text{m s}^{-1}$	[1] sub [1] ans
7(a)(iii)	Upon the closing of the switch, the electric field is established almost instantaneously, causing all electrons to move in the circuit. Although the drift velocity of the electrons is low, there is a large number of electrons in the circuit moving together once the switch is closed. Therefore, current in the circuit flows almost immediately.	[1] [1]
7(b)	Potential difference across wire X = 2.8 V Potential difference across NTC thermistor = 5.7 V $9.0 = 2.8 + 5.7 + 0.52r$ $r = 0.96\ \Omega$	[1] correct p.d for both wire X and NTC thermistor [1] ans
8(a)(i)	A = 94, Z = 35	[1] both correct